Photosynthesis

Overview

Photosynthesis is the process by which plants, algae and some bacteria turn light energy into chemical energy. The overall result is glucose and oxygen, made from carbon dioxide and water using light.

Where it happens

In plants, photosynthesis takes place in chloroplasts, which are found mainly in leaf cells. Chloroplasts contain chlorophyll, the green pigment that captures light. Chlorophyll absorbs mostly green light, which is why plants look green.

Light-dependent reactions

The first stage happens in the thylakoid membranes. Chlorophyll absorbs sunlight and uses its energy to split water. This releases oxygen as a by-product and produces two energy carriers, ATP and NADPH.

The Calvin cycle

The second stage happens in the stroma of the chloroplast and does not need light directly. The plant uses the ATP and NADPH from the first stage to turn carbon dioxide into glucose. The cycle repeats many times to build each sugar molecule.

What affects the rate

Three main factors change how fast photosynthesis runs: light intensity, carbon dioxide concentration and temperature. If any one of them is in short supply, it limits the whole process.